Report for Biostatistik

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1 Introduction

In this report a research question will be covered from the study "Explainable AI techniques with application to NBA gameplay prediction", namely: Has there been an evolution in game-play style from NBA teams through the years? This is helpful to analyse and understand the evolution of basketball, in order to further deepen our insights of the game. These differences will be analysed from a statistical standpoint and will be depicted graphically to further enhance comprehension.

2 Methods

2.1 Study design

The original study uses open-source NBA data from 1980 to 2019, collected from the website http://basketball.realgm.com/. The data includes seasonal and monthly game-play statistics for NBA teams. The data was scraped, joined, and aggregated from the website. The dataset used in this report is a simpler simulated version of the original one, including 1230 data points for the descriptive analysis.

This study is observational rather than experimental, it was not intervened with the variables at all. Already existing NBA game-play data was analyzed to identify trends and predict outcomes. This study is also retrospective because it analyzes historical data from 1980 to 2020. New data was not collected or teams were not followed prospectively.

The descriptive analysis uses seasonal data from 1980 to 2020, providing 40 years of data (40 time points). Also the report has repeated measurements of the same NBA teams across different seasons and months. Each team's performance is tracked over multiple time points.

Therefore we can say that this study design is a simulated retrospective observational study that uses historical NBA data to analyze game-play trends. The study involves repeated measurements of the same teams over multiple seasons and months, with a focus on descriptive analysis.

2.2 Data

The simulated data set is designed to recreate NBA team statistics across two distinct eras: 1980–2000 (Era 1: Defensive Play-style) and 2000–2020 (Era 2: Offensive Play-style). The data set is structured as follows:

- **Team**: A categorical variable representing the team name. There are 30 unique teams.
- **Year**: A numerical variable representing the year of the season, ranges from 1980 to 2020.
- **FGM** (Field Goals Made): A numerical variable representing the team's average efficiency in scoring from the field (excluding free throws and three-pointers).
- **3PM** (Three-Point Made): A numerical variable representing the average three-points made by the team in the given season.
- **3PA** (Three-Point Attempted): A numerical variable representing the average three-points attempted by the team in the given season.
- **FTM** (Free Throws Made): A numerical variable representing the average free throws made by the team in the given season.
- **FTA** (Free Throws Attempted): A numerical variable representing the average free throws attempted by the team in the given season.
- **TOV** (Turnovers): A numerical variable representing the average number of turnovers per game by a team in a given season.
- **DRB** (defensive Rebounds): A numerical variable representing the average defensive rebounds per game by a team in a given season.
- **PPG** (Points Per Game): A numerical variable representing the average points scored per game by a team in a given season.

The data set contains repeated measurements for each team across multiple seasons. Each team has 40 measurements for each season year, making it a longitudinal data set, this data was also afterwards standarized as suggested on the original study.

The Normal Distribution was used here for simulating all continuous variables while a set of repetitions was used for simulating teams and years.

The simulation works with independence between variables, except for the variable Year, which is correlated with all other variables. And, in reality, some other variables may also be correlated (e.g., higher 3PA might lead to higher 3PM). Also for this descriptive analysis no dependent variable was simulated/needed.

2.3 Statistical Analysis

In this descriptive analysis, different statistical methods were implemented to study how NBA game-play has changed from the 1980s to the 2010s. Principal Component Analysis (PCA) was used in addition to clustering techniques (K-means and DBSCAN) to first reduce the number

of variables and create 2D or 3D plots that show how the clusters differ, which grouped seasons into different eras based on game-play styles, such as the defensive-focused 1980s versus the three-point-heavy 2010s.

The goal of these methods is to give a clear picture of how NBA game-play has shifted over time. By using clustering and visualizations, we can identify and understand distinct eras of play styles and show how specific stats have changed.

3 Results

After simulating our data, we proceed to standardize it and preform PCA, furthermore we compare K-Means and DBSCAN clustering techniques based on the best fit.

Table 1: Importance of components

	PC1	PC2	PC3	PC4
Standard deviation	1.7856	1.0182	0.9978	0.9469
Proportion of Variance	0.3986	0.1296	0.1245	0.1121
Cumulative Proportion	0.3986	0.5282	0.6526	0.7647

It can be observed in **table 1**, that the first 3 PC only explain around 65% of the total variance, therefore this study acts only as a reference for trend visualization and not as the golden standard rule of NBA game play evolution.

Next a DBSCAN technique was used with a epsilon of 2.4, this technique preformed poorly since it grouped almost all data points into just one cluster as seen in **table 2**.

Table 2: DBSCAN clustering results with epsilon of 2.4

Cluster	Cluster size
0	4
1	1226

Next, when using a epsilon value of 4.5, we find that this technique preforms even worst than before and it cannot be used for grouping purposes for it grouped all points in one cluster, as seen in **table 3**.

Table 3: DBSCAN clustering results with epsilon of 4.5

Cluster	Cluster size
0	0
1	1230

Since DBSCAN does not have a good fit for the data set, it was decided to continue with the K-means clustering technique, which preformed the best with 2 clusters given that it is the highest Silhouette score as can be observed in **table 4**.

Table 4: K-means clustering results.

Cluster:	Silhouette Score:
2	0.2453919
3	0.1355320
4	0.1254295
5	0.1168668
6	0.1122934
7	0.1124074
8	0.1120860
9	0.1127173

3.1 Descriptive analysis and results.

After deciding the best technique to use, we continue to preform a graphic interpretation from the clustering techniques together with the PCA, in order to finally identify whether there were changes or not in game-play styles.

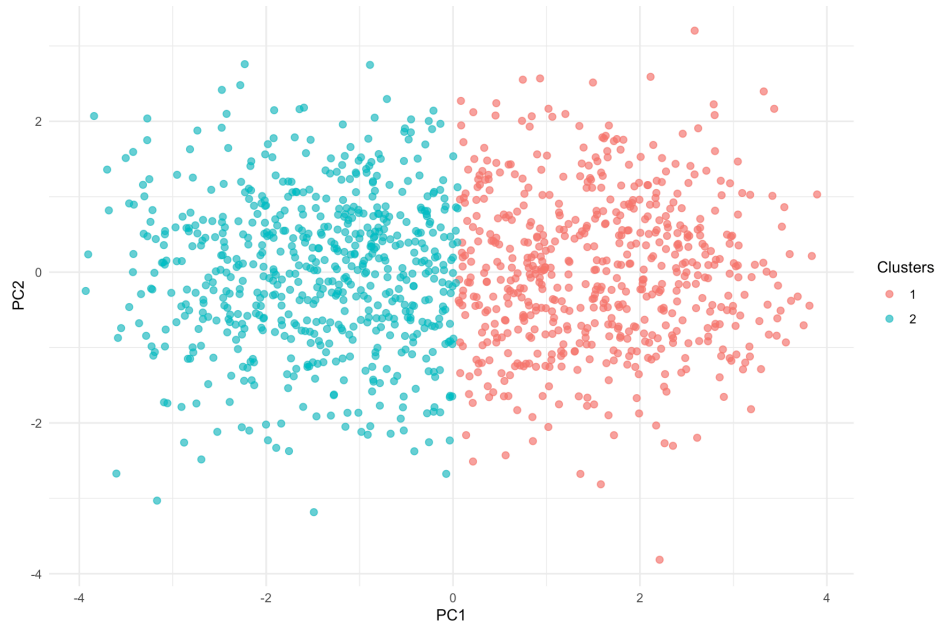

Figure 1: K-means Clustering of NBA Team Data

We can clearly see in **Figure 1** that there is a difference in game-play styles between the 2 eras, this can be appreciated, given that the 2 principal components have condensed the high dimensionallity of the data set and the K-means clustering technique has grouped these data points into the different dynasties of Basketball.

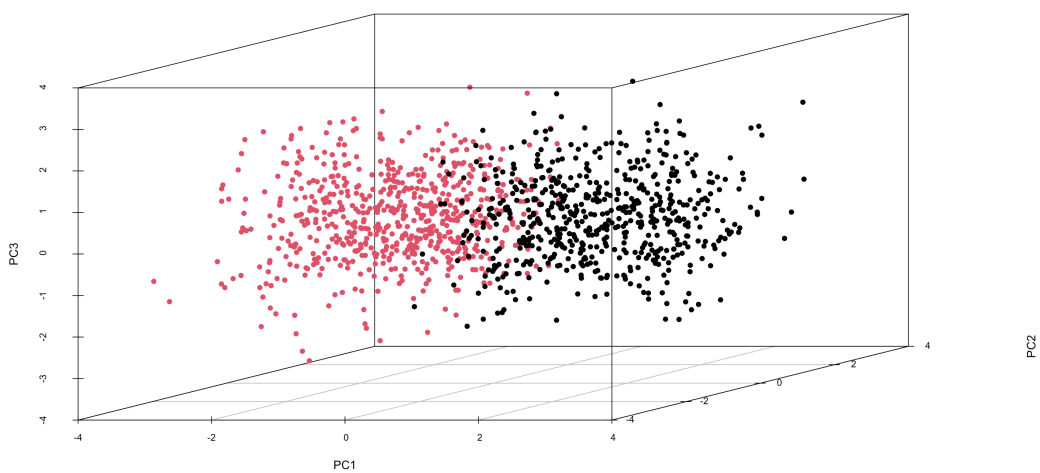

Figure 2: K-means Clustering (3D)

Now when using 3 PC, we can observe in **Figure 2** that the differences can be even better appreciated, this is also because, using 3 components explains the variance of the data way more than just 2. The plot shows that: 1) the game-play varies across the different clusters found; and 2) the game-play gradually changes when the time frame is altered.

To conclude, a trend analysis was conducted in order to analyse the changes in averages of each attribute over the years, this will help to better visualize the evolution of this sport.

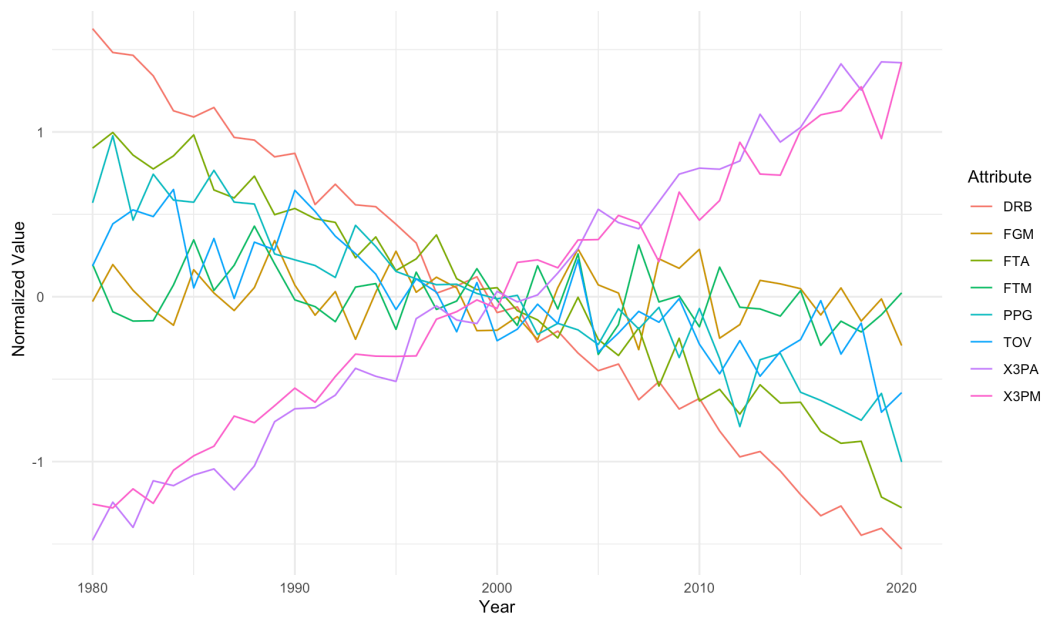

Figure 3: Trend Analysis of game-play attributes.

All 8 attributes are shown on a single graph as all values have been standardized. From the standpoint of an NBA fan, the trend is really interesting, since for every case, a few observations can be made along with several potential theories, such as the following:

- Over the course of four decades, there has been a consistent increase in 3-point attempts and 3-pointers made;
- A continuous decrease in defensive rebounds and a continuous decrease in free throw attempts and free throws made.

According to **figure 3** above, the NBA's style varies greatly depending on the year. One interesting finding is that a player who could have been successful in 1989 would not be able to translate his skill set into success in 2019.

4 Conclusions

Obvious changes in play-styles can be obsteved by examining patterns and grouping seasons into discrete periods. For example, the defensive-oriented basketball of the 1980s and 1990s contrasts with the faster-paced, attack heavy style of the 2000s and 2010's. The league's shift to a more offensive style of play is seen in the notable rise in three-point attempts and scoring, as well as the decline in rebounds and blocks (see **figure 3**).

With the help of statistical summaries and visualizations, the use of trend analysis, PCA, and clustering, a straightforward and intuitive understanding of these changes can be recognized. All things considered, the descriptive analysis offers valuable insights on how the NBA's game-play has transformed over four decades, reflecting big changes in strategies, player skills, and league trends, which can inform strategic decisions, be used by coaches, analyst to adapt to evolving trends, or simply fans to compare team game styles within specif eras.